Experiment 3: Demonstrate the working of the decision tree based ID3algorithm. Use an appropriate data setfor building the decision tree and apply this knowledge to classify a new sample.

1.student plaement prediction dataset

CGPA	**Aptitude Score**	**Communication Score**	**Placement**
9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

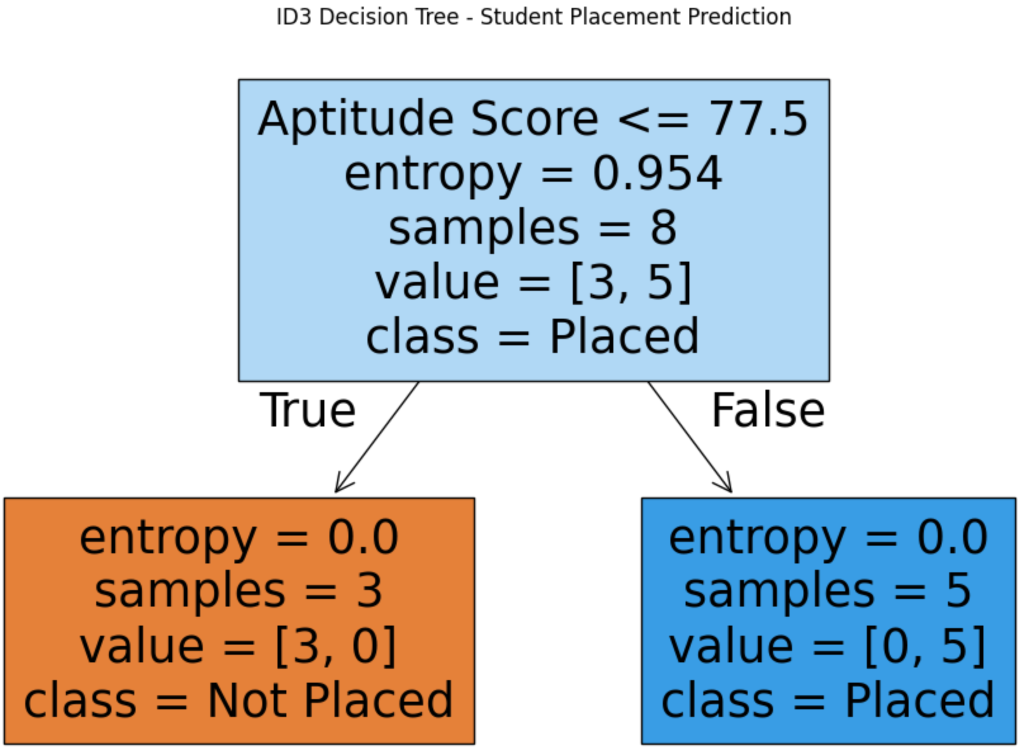
CODE:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import accuracy_score

import matplotlib.pyplot as plt

data = pd.read_csv("student plaement prediction .csv")

print("Dataset:")

print(data)
```

```
X = data.drop("Placement", axis=1)
y = data["Placement"]
label_encoder = LabelEncoder()
y = label_encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nAccuracy:", accuracy_score(y_test, y_pred))
new_student = [[8.7, 84, 85]]
prediction = model.predict(new_student)
print("\nNew Student:")
print("CGPA = 8.7")
print("Aptitude Score = 84")
print("Communication Score = 85")

print("Prediction:",
      label_encoder.inverse_transform(prediction)[0])
plt.figure(figsize=(12, 8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=label_encoder.classes_,
    filled=True
```

```
)
plt.title("ID3 Decision Tree - Student Placement Prediction")
plt.show()
```

OUTPUT:



2. loan approval prediction dataset

Income (₹ Lakhs)	**Credit Score**	**Loan Amount (₹ Lakhs)**	**Loan Approved**
8	780	5	Yes
7	760	4	Yes
6	720	6	Yes
4	620	8	No
3	580	10	No
9	800	4	Yes
5	650	7	No
8	770	5	Yes
4	600	9	No
7	740	6	Yes

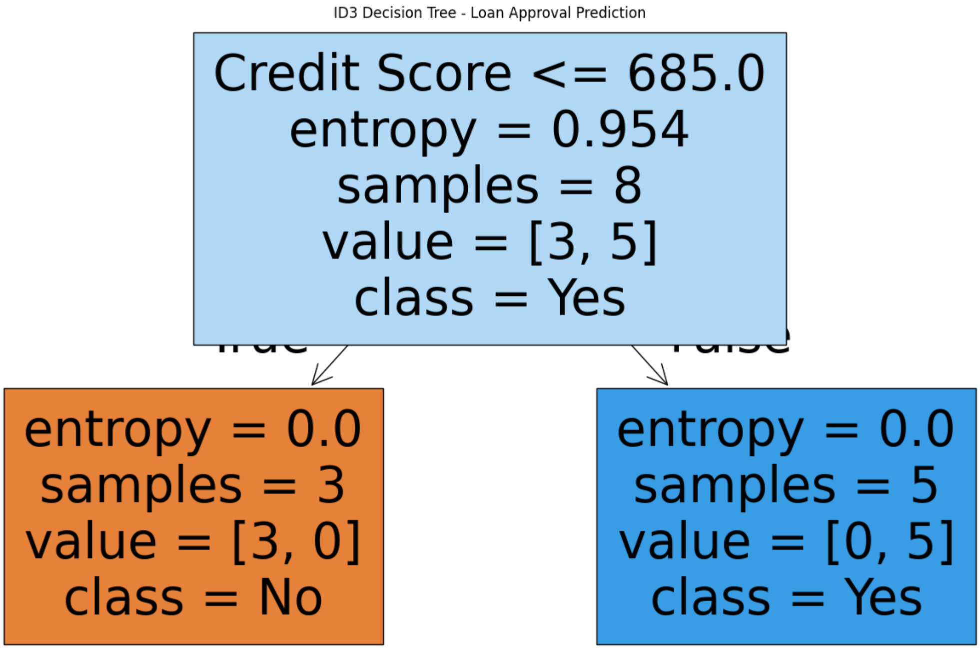
Code:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import accuracy_score

import matplotlib.pyplot as plt

data = pd.read_csv("loan approval prediction .csv")

print("Dataset:")

print(data)

data = data.dropna()

target = "Loan Approved"

X = data.drop(target, axis=1)
```

```
y = data[target]
X = pd.get_dummies(X)
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nAccuracy:", accuracy_score(y_test, y_pred))
new_sample = X_test.iloc[[0]]
prediction = model.predict(new_sample)
print("\nNew Loan Application:")
print(new_sample)
print("\nPrediction:",
      encoder.inverse_transform(prediction)[0])
plt.figure(figsize=(18, 10))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=encoder.classes_,
    filled=True
)
```

```
plt.title("ID3 Decision Tree - Loan Approval Prediction")
plt.show()
```

Output:



3. **disease diagnosis**

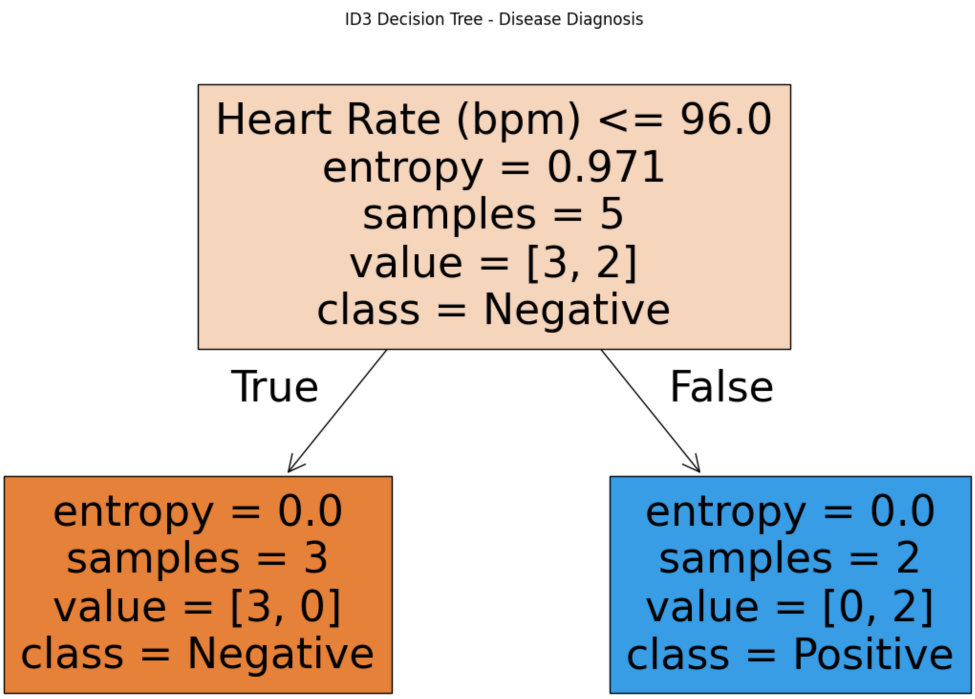
Temperature (°C)	**Heart Rate (bpm)**	**Oxygen Level (%)**	**Disease**
39	110	94	Positive
38.5	108	95	Positive
37	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
data = pd.read_csv("disease diagnosis .csv") # Corrected filename
print("Dataset:")
print(data)
data = data.dropna()
target = "Disease"
X = data.drop(target, axis=1)
y = data[target]
X = pd.get_dummies(X)
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
```

```
new_patient = X_test.iloc[[0]]
prediction = model.predict(new_patient)
print("\nNew Patient:")
print(new_patient)
print("\nPredicted Disease:",
      encoder.inverse_transform(prediction)[0])
plt.figure(figsize=(16, 10))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=encoder.classes_,
    filled=True
)
plt.title("ID3 Decision Tree - Disease Diagnosis")
plt.show()
```

Output:



4. Employee Promotion Prediction dataset

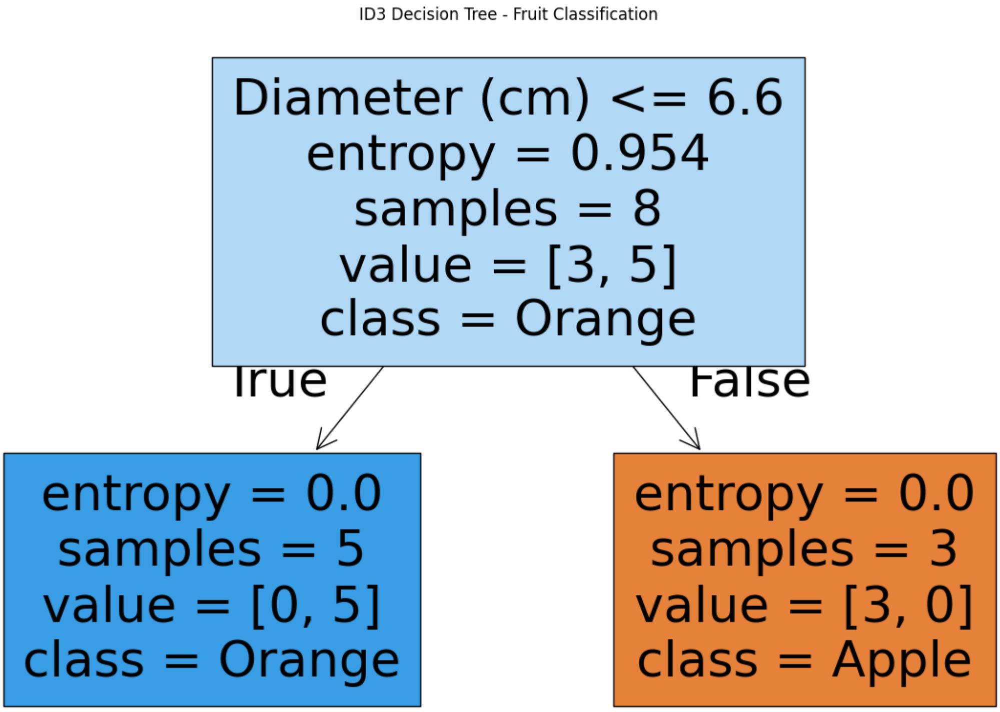
Experience (Years)	Performance Score	Training Hours	Promotion
10	95	50	Yes
8	90	45	Yes
7	88	40	Yes
3	65	20	No
2	60	18	No
9	93	48	Yes
4	70	25	No
8	89	42	Yes
2	58	15	No
6	85	38	Yes

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
data = pd.read_csv("Fruit Classification .csv")
print("Dataset:")
print(data)
data = data.dropna()
target = "Fruit"
X = data.drop(target, axis=1)
y = data[target]
X = pd.get_dummies(X)
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nAccuracy:",
```

```
    accuracy_score(y_test, y_pred))
new_fruit = X_test.iloc[[0]]

prediction = model.predict(new_fruit)

print("\nNew Fruit:")
print(new_fruit)

print("\nPredicted Fruit:",
    encoder.inverse_transform(prediction)[0])
plt.figure(figsize=(16, 10))

plot_tree(
   model,
   feature_names=X.columns,
   class_names=encoder.classes_,
   filled=True
)

plt.title("ID3 Decision Tree - Fruit Classification")
plt.show()
```

Output:



5. Fruit Classification dataset

Weight (g)	**Diameter (cm)**	**Sweetness (%)**	**Fruit**
150	7.5	90	Apple
160	7.8	88	Apple
140	7.2	91	Apple
120	5.5	70	Orange
125	5.8	72	Orange
130	6	74	Orange
155	7.6	89	Apple

118	5.4	69	Orange
148	7.4	92	Apple
122	5.7	71	Orange

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
data = pd.read_csv("Employee Promotion Prediction .csv")
print("Dataset:")
print(data)
data = data.dropna()
target = "Promotion"
X = data.drop(target, axis=1)
y = data[target]
X = pd.get_dummies(X)
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
```

```
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
new_employee = X_test.iloc[[0]]
prediction = model.predict(new_employee)
print("\nNew Employee:")
print(new_employee)
print("\nPromotion Prediction:",
      encoder.inverse_transform(prediction)[0])
plt.figure(figsize=(18, 10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=encoder.classes_,
    filled=True
)

plt.title("ID3 Decision Tree - Employee Promotion Prediction")
plt.show()
```

Output:

ID3 Decision Tree - Employee Promotion Prediction